Qunsheng Zhang

CURRICULUM VITAE

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Research Interests Safety-certified control of multi-agent systems; control-barrier and Lyapunov certificates; multi-agent reinforcement learning; runtime verification and shielding; trustworthy agentic AI; embodied multi-agent systems

PROFILE

Mathematician with an MSc from Southeast University and **two first-author papers in JCR Q1 journals** on predefined-time and fixed-time coordination of nonlinear multi-agent systems. My work combines Lyapunov and barrier arguments, sliding-mode and prescribed-performance control, optimisation and reinforcement learning. In 2025–2026 I **led the security-operations LLM initiative** at Bank of Communications' head-office data center, building auditable multi-agent systems under production and compliance constraints. I aim to make embodied multi-agent systems both capable and certifiably safe: learned policies with enforceable guarantees, runtime shields, and evidence a supervisor can audit.

EDUCATION

MSc in Mathematics
2022 – 2025

Southeast University, Nanjing — Weighted avg. **85.81/100**

- Coursework: Functional Differential Equations (91), Mathematical Methods in Information Science (93), Nonlinear Control Systems, Convex Optimisation, Stochastic Analysis.
- Research areas: multi-agent systems; reinforcement learning; sliding-mode control; distributed optimisation.

BSc in Mathematics and Applied Mathematics
2018 – 2022

University of Shanghai for Science and Technology, Shanghai — GPA **3.79/4.50** | **top 2%**

- Coursework: Mathematical Analysis (97), Advanced Algebra (99), Complex Analysis (97), Ordinary Differential Equations, Operations Research (92), Numerical Analysis and MATLAB (91).
- **Shanghai Outstanding Graduate; Second-Class Academic Scholarship.**

PUBLICATIONS

Peer-reviewed

1. **Q. Zhang**, J. Hu, J. Cao, T. Yu. "Optimized predefined-time control for high-order nonlinear MASs via ICA and reinforcement learning." *ISA Transactions*, vol. 176 (2026), pp. 436–448. JCR Q1. doi:10.1016/j.isatra.2026.06.029
2. **Q. Zhang**, J. Hu, J. Cao. "Fixed-time synchronization of nonlinear coupling MNNs with time delay via aperiodically intermittent sliding mode control." *Communications in Nonlinear Science and Numerical Simulation*, vol. 150 (2025), art. 108995. JCR Q1. doi:10.1016/j.cnsns.2025.108995

Under review

3. **Q. Zhang**. "VERA: Root-Domain Evidence Quorums for Carrier Integrity in Tool-Calling Agents." *AAAI 2027*, under review. CCF-A.

RESEARCH EXPERIENCE

Resilient Cooperative Control and Reinforcement Learning
2024 – 2025

NSFC General Program (Grant No. 62573116) | Southeast University | **Key research contributor**

Supervisors: Associate Prof. Jianqiang Hu; Academician Jinde Cao

Background: Cyberattacks and uncertain dynamics can disrupt coordination in multi-agent and new-type power systems.

- Designed predefined-time and event-triggered optimal cooperative control for unknown high-order nonlinear MASs using RL and intelligent optimisation, improving convergence accuracy and reducing communication energy in dynamic settings.
- **First author** of related work published in *ISA Transactions*(JCR Q1), a leading control and systems science journal.

Distributed Secure Cooperative Control for Complex Cyber-Physical Networks
2022 – 2024

NSFC General Program (Grant No. 62073079) | Southeast University | **Key research contributor**

Background: Networked control over wireless links faces attacks, delays, packet loss and resource constraints.

- Designed aperiodically intermittent sliding-mode control for nonlinearly coupled memristive neural networks with time delays; derived fixed-time synchronisation criteria and settling-time bounds valid for any initial condition.
- Analysed how aperiodic control and communication schedules affect certified convergence; validated on small-world network examples; **first-author** paper in *CNSNS* (JCR Q1).

INDUSTRY EXPERIENCE

Security Automation & LLM Implementation
2025.07 – 2026.07

Bank of Communications, Head Office Data Center | Shanghai

- **Led** the production integration of LLMs into the 360 Next-Gen SOC, introducing AI-assisted alert analysis and attack-chain reconstruction alongside incident-response playbooks.
- Engineered automated triage and remediation workflows for security telemetry from Splunk, Colasoft and Alibaba Cloud; increased the **alert remediation automation rate to 97%**, reducing manual handling and human error.
- Designed L1/L2 agent routing, MCP tool calls and hybrid retrieval across vector search, BM25 and IOC matching; used multi-source evidence to support faster detection and response (MTTD and MTTR).
- Built auditable controls including MITRE ATT&CK mapping, confidence calibration, human review and execution-boundary monitors to constrain tool-calling agents under adversarial inputs.

HONORS & AWARDS

- **National recommendation for Outstanding Master's Thesis**, Southeast University — 2025
- Oral presentation, CSSCC (Chinese Conference on System Science and Control) — 2024
- National 2nd Prize, 20th China Graduate Mathematical Contest in Modeling (Huawei Cup) — 2023
- **Shanghai Outstanding Graduate**; Second-Class Academic Scholarship, Southeast University — 2022
- National 3rd Prize, 19th China Graduate Mathematical Contest in Modeling (Huawei Cup) — 2022
- National 2nd Prize, 13th Chinese Mathematics Competitions (CMC), Mathematics Majors Category — 2021
- Academic Excellence Scholarship, University of Shanghai for Science and Technology — 2019 – 2021

SKILLS & LANGUAGES

Theory	Lyapunov stability; fixed-time and predefined-time convergence; sliding-mode control; networked and cyber-physical control; nonlinear multi-agent systems; functional differential equations; convex optimisation; matrix analysis; probability and stochastic analysis.
Proof techniques	Lyapunov and Lyapunov–Krasovskii functionals for time-delay systems; fixed-time and predefined-time stability lemmas; sliding-mode surface design; projection-based adaptation; graph-theoretic analysis of multi-agent networks.
Computing	Python (NumPy, SciPy, PyTorch), MATLAB/Simulink, C/C++, LaTeX, Linux, Git; numerical simulation, algorithm development and reproducible research.
Languages	Chinese (native); English (professional working proficiency — CET-6; daily technical writing and presentation).

TEACHING EXPERIENCE

Teaching Assistant, Complex Systems and Equations
2022 – 2025

Southeast University | School of Mathematics

- Graded weekly assignments and provided written feedback; led tutorial sessions on complex systems and differential equations.

Teaching Assistant, School of Science
2021 – 2022

University of Shanghai for Science and Technology | School of Science

- Led tutorial sessions for Mathematical Analysis, Ordinary Differential Equations, Probability Theory and Abstract Algebra; prepared and presented worked solutions to weekly problem sets.

REFERENCES

Letters available on request

Jinde Cao
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Southeast University, Nanjing, China
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M.Sc. co-supervisor; IEEE Fellow; Member, Academia Europaea

Jianqiang Hu
Associate Professor, School of Mathematics
Southeast University, Nanjing, China
jqhu@seu.edu.cn
M.Sc. supervisor, 2022–2025; co-author on both papers

Jianhua Hu
Associate Professor, School of Science
University of Shanghai for Science and Technology, Shanghai
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B.Sc. thesis advisor; tutor in Linear and Abstract Algebra